

T3 intermediate-EFT one-loop RGE term comparison: EFT_1 after F

This stage has active field content $\text{SM} + S_1 + S_2$. The report combines the renormalisable RGE running with the dimension-five $\psi^2\phi^2$ Wilson-coefficient running generated after integrating out F . Throughout, $16\pi^2 \mu dX/d\mu = \beta_X^{(1)}$. For the Wilson sector only the tree-generated stage-1 coefficient $C^{(0)}$ is inserted into the one-loop anomalous dimension; the one-loop threshold piece is retained as a boundary term and is not run again at this order.

Notation key

symbol	interaction / definition
g_1	$D_\mu \supset ig_1 Y B_\mu$
g_2	$D_\mu \supset ig_2 T^A W_\mu^A$
g_3	$D_\mu \supset ig_3 t^A G_\mu^A$
Y_u	$-\bar{Q}Y_u\tilde{H}u + \text{h.c.}$
Y_d	$-\bar{Q}Y_dHd + \text{h.c.}$
Y_e	$-\bar{L}Y_eHe + \text{h.c.}$
y_1	$LF S_1 + \text{h.c.}$
y_2	$LF S_2 + \text{h.c.}$
M_F	$-\bar{F}M_F F$
T	$\text{Tr}(Y_e^\dagger Y_e + 3Y_u^\dagger Y_u + 3Y_d^\dagger Y_d)$
$T_\nu^{(1)}$	$\text{Tr}(y_1^\dagger y_1)$
$T_\nu^{(2)}$	$\text{Tr}(y_2^\dagger y_2)$
T_ν	$T_\nu^{(1)} + T_\nu^{(2)}$
m_1^2	$m_1^2 S_1^\dagger S_1$
m_2^2	$m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2}\lambda_1 (H^\dagger H)^2$
$\lambda_{S_1}^{(1)}$	$\frac{1}{2}\lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$
$\lambda_{S_2}^{(1)}$	$\frac{1}{2}\lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$
$\lambda_{HS_1}^{(1)}$	$\lambda_{HS_1}^{(1)} (H^\dagger H)(S_1^\dagger S_1)$
$\lambda_{HS_2}^{(1)}$	$\lambda_{HS_2}^{(1)} (H^\dagger H)(S_2^\dagger S_2)$
$\lambda_{HS_1}^{(A)}$	$\lambda_{HS_1}^{(A)} (H^\dagger T^A H)(S_1^\dagger T^A S_1)$
$\lambda_{HS_2}^{(A)}$	$\lambda_{HS_2}^{(A)} (H^\dagger T^A H)(S_2^\dagger T^A S_2)$
λ_5	$\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$
$\lambda_{12}^{(1)}$	$\lambda_{12}^{(1)} (S_1^\dagger S_1)(S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$\lambda_{12}^{(A)} (S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	$\lambda_{12}^{(\times)} (S_1^\dagger S_2)(S_2^\dagger S_1)$
$\lambda_{S_1}^{(A)}$	$\lambda_{S_1}^{(A)} (S_1^\dagger T^A S_1)(S_1^\dagger T^A S_1)$
$\lambda_{S_2}^{(A)}$	$\lambda_{S_2}^{(A)} (S_2^\dagger T^A S_2)(S_2^\dagger T^A S_2)$
α	$Y_{S_1} = \alpha/2, Y_F = (\alpha+1)/2, Y_{S_2} = (\alpha+2)/2$

$C_2(d)$	$(d^2-1)/4$
$T(d)$	$d(d^2-1)/12$
$C_{LLS_1S_2}^{ij;AB}$	$L_iL_jS_1^AS_2^B$
C_{ijab}	$L_iL_j\phi_a\phi_b$
C_5^{ij}	$(L_i^TCL_j)HH$
R_W	$16\pi^2\beta_{C_5}\supset R_W\lambda_5C_{LLS_1S_2}$

Renormalisable EFT1 couplings

$\beta_{g_1}^{(1)}$

Quantity: g_1

$16\pi^2\,\mu\frac{dg_1}{d\mu}=\beta_{g_1}^{(1)}$

Column Representative term structure	
T1	$\frac{47g_1^3}{6}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{47g_1^3}{6}$

$\beta_{g_2}^{(1)}$
Quantity: g_2

$$16\pi^2\,\mu\frac{dg_2}{d\mu}=\beta_{g_2}^{(1)}$$

Column	Representative term structure
T1	$-\frac{11g_2^3}{6}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{11g_2^3}{6}$

$\beta_{g_3}^{(1)}$
Quantity: g_3

$$16\pi^2\mu\frac{dg_3}{d\mu}=\beta_{g_3}^{(1)}$$

Column	Representative term structure
T1	$-7g_3^3$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-7g_3^3$

$$\beta_{Y_u}^{(1)}$$

Quantity: Y_u

$$16\pi^2\,\mu\frac{dY_u}{d\mu}=\beta_{Y_u}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$-3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_u)_{ij}$
T2	$+ \mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_u)_{ij}$
T3	$+3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_u)_{ij}$	$+ \mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_u)_{ij}$	$+3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_u)_{ij}$

Quantity: Y_u

Terms 2 of 3

Column	Representative term structure
T1	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$
T2	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$
T3	$-8g_3^2\ (Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_u)_{ij}$	$-8g_3^2\ (Y_u)_{ij}$

Quantity: Y_u

Terms 3 of 3

Column	Representative term structure
T1	$-\frac{17}{12}g_1^2(Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$

$\beta_{Y_d}^{(1)}$

Quantity: Y_d

$16\pi^2\mu\frac{dY_d}{d\mu}=\beta_{Y_d}^{(1)}$

Terms 1 of 3

Column	Representative term structure
T1	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\ (Y_d)_{ij}$
T2	$+3\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\ (Y_d)_{ij}$
T3	$+3\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\ (Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\ (Y_d)_{ij}$	$+3\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\ (Y_d)_{ij}$	$+3\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\ (Y_d)_{ij}$

Quantity: Y_d

Terms 2 of 3

Column	Representative term structure
T1	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$
T2	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$
T3	$-8g_3^2\ (Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$

Quantity: Y_d

Terms 3 of 3

Column	Representative term structure
T1	$-\frac{5}{12}g_1^2(Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$

$\beta_{Y_e}^{(1)}$
Quantity: Y_e

$16\pi^2\mu\frac{dY_e}{d\mu}=\beta_{Y_e}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$-3\mathrm{Tr}\Big(Y_d\;(Y_d)^\dagger\Big)\;(Y_e)_{ij}$
T2	$+3\mathrm{Tr}\Big(Y_u\;(Y_u)^\dagger\Big)\;(Y_e)_{ij}$
T3	$+\mathrm{Tr}\Big(Y_e\;(Y_e)^\dagger\Big)\;(Y_e)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$3\mathrm{Tr}\Big(Y_d\;(Y_d)^\dagger\Big)\;(Y_e)_{ij}$	$+3\mathrm{Tr}\Big(Y_u\;(Y_u)^\dagger\Big)\;(Y_e)_{ij}$	$+\mathrm{Tr}\Big(Y_e\;(Y_e)^\dagger\Big)\;(Y_e)_{ij}$

Quantity: Y_e

Terms 2 of 2

Column	Representative term structure
T1	$+\frac{3}{2}\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j}$
T2	$-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$
T3	$-\frac{15}{4} g_1^2\left(Y_e\right)_{i j}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{3}{2}\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j}$	$-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$	$-\frac{15}{4} g_1^2\left(Y_e\right)_{i j}$

$\beta_{m_1^2}^{(1)}$

Quantity: m_1^2

$16\pi^2\mu\frac{dm_1^2}{d\mu}=\beta_{m_1^2}^{(1)}$

Column	Representative term structure
T1	$-12g_2^2m_1^2$
T2	$+m_2^2(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$
T3	$+m_1^2(8\lambda_{S_1}^{(1)}+4\lambda_{S_1}^{(A)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-12g_2^2m_1^2$	$+m_2^2(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$	$+m_1^2(8\lambda_{S_1}^{(1)}+4\lambda_{S_1}^{(A)})$

$\beta_{m_2^2}^{(1)}$

Quantity: m_2^2

$16\pi^2\mu\frac{dm_2^2}{d\mu}=\beta_{m_2^2}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$-12g_2^2m_2^2$
T2	$-6g_1^2m_2^2$
T3	$+m_1^2(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-12g_2^2m_2^2$	$-6g_1^2m_2^2$	$+m_1^2(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$

Quantity: m_2^2

Terms 2 of 2

Column	Representative term structure
T1	$+m_2^2(8\lambda_{S_2}^{(1)}+4\lambda_{S_2}^{(A)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+m_2^2(8\lambda_{S_2}^{(1)}+4\lambda_{S_2}^{(A)})$

$\beta_{\lambda_1}^{(1)}$
Quantity: λ_1

$$16\pi^2\,\mu\,\frac{d\lambda_1}{d\mu}=\beta_{\lambda_1}^{(1)}$$

Terms 1 of 5

Column	Representative term structure
T1	$\lambda_1(12\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+4\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+12\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$
T2	$-12\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\,Y_d\,(Y_d)^\dagger\big)$
T3	$-4\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\,Y_e\,(Y_e)^\dagger\big)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_1(12\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+4\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+12\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$	$-12\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\,Y_d\,(Y_d)^\dagger\big)$	$-4\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\,Y_e\,(Y_e)^\dagger\big)$

Quantity: λ_1

Terms 2 of 5

Column	Representative term structure
T1	$-12\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\,Y_u\,(Y_u)^\dagger\Big)$
T2	$+4\lambda_5\,(\lambda_5)^*$
T3	$+ \frac{9g_2^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-12\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\,Y_u\,(Y_u)^\dagger\Big)$	$+4\lambda_5\,(\lambda_5)^*$	$+ \frac{9g_2^4}{4}$

Quantity: λ_1

Terms 3 of 5

Column	Representative term structure
T1	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$
T2	$-9g_2^2\lambda_1$
T3	$+\frac{3g_1^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$

Terms 4 of 5

Column	Representative term structure
T1	$+12\lambda_1^2$
T2	$+6\lambda_{HS_1}^{(1)2}$
T3	$+3\lambda_{HS_1}^{(A)2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+12\lambda_1^2$	$+6\lambda_{HS_1}^{(1)2}$	$+3\lambda_{HS_1}^{(A)2}$

Quantity: λ_1

Terms 5 of 5

Column	Representative term structure
T1	$+6\lambda_{HS_2}^{(1) \ 2}$
T2	$+3\lambda_{HS_2}^{(A) \ 2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+6\lambda_{HS_2}^{(1) \ 2}$	$+3\lambda_{HS_2}^{(A) \ 2}$

$\beta_{\lambda_{S_1}^{(1)}}^{(1)}$

Quantity: $\lambda_{S_1}^{(1)}$

$16\pi^2\mu\frac{d\lambda_{S_1}^{(1)}}{d\mu}=\beta_{\lambda_{S_1}^{(1)}}^{(1)}$

Terms 1 of 4

Column	Representative term structure
T1	$21g_2^4$
T2	$+g_2^2(-24\lambda_{S_1}^{(1)}-12\lambda_{S_1}^{(A)})$
T3	$+6\lambda_{12}^{(1)2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$21g_2^4$	$+g_2^2(-24\lambda_{S_1}^{(1)}-12\lambda_{S_1}^{(A)})$	$+6\lambda_{12}^{(1)2}$

Quantity: $\lambda_{S_1}^{(1)}$

Terms 2 of 4

Column	Representative term structure
T1	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$
T2	$+6\lambda_{12}^{(A)2}$
T3	$+6\lambda_{12}^{(\times)2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	$+6\lambda_{12}^{(A)2}$	$+6\lambda_{12}^{(\times)2}$

Quantity: $\lambda_{S_1}^{(1)}$

Terms 3 of 4

Column	Representative term structure
T1	$+4\lambda_{HS_1}^{(1) \ 2}$
T2	$+\frac{3\lambda_{HS_1}^{(A) \ 2}}{2}$
T3	$+14\lambda_{S_1}^{(1) \ 2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{HS_1}^{(1) \ 2}$	$+\frac{3\lambda_{HS_1}^{(A) \ 2}}{2}$	$+14\lambda_{S_1}^{(1) \ 2}$

Quantity: $\lambda_{S_1}^{(1)}$

Terms 4 of 4

Column	Representative term structure
T1	$+14\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$
T2	$+26\lambda_{S_1}^{(A)2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+14\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$	$+26\lambda_{S_1}^{(A)2}$

$$\beta_{\lambda_{S_2}^{(1)}}^{(1)}$$

Quantity: $\lambda_{S_2}^{(1)}$

$$16\pi^2\,\mu\frac{d\lambda_{S_2}^{(1)}}{d\mu}=\beta_{\lambda_{S_2}}^{(1)}$$

Terms 1 of 5

Column	Representative term structure	
T1	$21g_2^4$	
T2	$+g_1^2\left(12g_2^2-12\lambda_{S_2}^{(1)}-6\lambda_{S_2}^{(A)}\right)$	
T3	$+g_2^2(-24\lambda_{S_2}^{(1)}-12\lambda_{S_2}^{(A)})$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$21g_2^4$	$+g_1^2\left(12g_2^2-12\lambda_{S_2}^{(1)}-6\lambda_{S_2}^{(A)}\right)$	$+g_2^2(-24\lambda_{S_2}^{(1)}-12\lambda_{S_2}^{(A)})$

Quantity: $\lambda_{S_2}^{(1)}$

Terms 2 of 5

Column	Representative term structure
T1	$+12g_1^4$
T2	$+6\lambda_{12}^{(1)2}$
T3	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+12g_1^4$	$+6\lambda_{12}^{(1)2}$	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$

Quantity: $\lambda_{S_2}^{(1)}$

Terms 3 of 5

Column	Representative term structure
T1	$+6\lambda_{12}^{(A)^2}$
T2	$+6\lambda_{12}^{(\times)^2}$
T3	$+4\lambda_{HS_2}^{(1)^2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+6\lambda_{12}^{(A)^2}$	$+6\lambda_{12}^{(\times)^2}$	$+4\lambda_{HS_2}^{(1)^2}$

Quantity: $\lambda_{S_2}^{(1)}$

Terms 4 of 5

Column	Representative term structure
T1	$+\frac{3\lambda_{HS_2}^{(A)^2}}{2}$
T2	$+14\lambda_{S_2}^{(1)^2}$
T3	$+14\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{3\lambda_{HS_2}^{(A)^2}}{2}$	$+14\lambda_{S_2}^{(1)^2}$	$+14\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$

Quantity: $\lambda_{S_2}^{(1)}$

Terms 5 of 5

Column	Representative term structure
T1	$+26\lambda_{S_2}^{(A)2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+26\lambda_{S_2}^{(A)2}$

$\beta_{\lambda_{HS_1}^{(1)}}^{(1)}$

Quantity: $\lambda_{HS_1}^{(1)}$

$16\pi^2\mu\frac{d\lambda_{HS_1}^{(1)}}{d\mu}=\beta_{\lambda_{HS_1}^{(1)}}^{(1)}$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda_{HS_1}^{(1)}\left(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)+4\lambda_{S_1}^{(A)}\right)$
T2	$+4\lambda_5\left(\lambda_5\right)^*$
T3	$+6g_2^4$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_{HS_1}^{(1)}\left(6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)+4\lambda_{S_1}^{(A)}\right)$	$+4\lambda_5\left(\lambda_5\right)^*$	$+6g_2^4$

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 2 of 4

Column	Representative term structure
T1	$-\frac{33g_2^2\lambda_{HS_1}^{(1)}}{2}$
T2	$-\frac{3g_1^2\lambda_{HS_1}^{(1)}}{2}$
T3	$+\lambda_{HS_2}^{(1)}(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{33g_2^2\lambda_{HS_1}^{(1)}}{2}$	$-\frac{3g_1^2\lambda_{HS_1}^{(1)}}{2}$	$+\lambda_{HS_2}^{(1)}(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 3 of 4

Column	Representative term structure
T1	$+6\lambda_1\lambda_{HS_1}^{(1)}$
T2	$+4\lambda_{HS_1}^{(1)^2}$
T3	$+8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$	$+8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 4 of 4

Column	Representative term structure
T1	$+6\lambda_{HS_1}^{(A)^2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+6\lambda_{HS_1}^{(A)^2}$

$\beta^{(1)}_{\lambda_{HS_2}^{(1)}}$

Quantity: $\lambda_{HS_2}^{(1)}$

$$16\pi^2\mu\frac{d\lambda_{HS_2}^{(1)}}{d\mu}=\beta^{(1)}_{\lambda_{HS_2}^{(1)}}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda_{HS_2}^{(1)}(6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big)+4\lambda_{S_2}^{(A)})$
T2	$+4\lambda_5\ (\lambda_5)^*$
T3	$+6g_2^4$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_{HS_2}^{(1)}(6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big)+4\lambda_{S_2}^{(A)})$	$+4\lambda_5\ (\lambda_5)^*$	$+6g_2^4$

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 2 of 4

Column	Representative term structure
T1	$-\frac{33g_2^2\lambda_{HS_2}^{(1)}}{2}$
T2	$+3g_1^4$
T3	$-\frac{15g_1^2\lambda_{HS_2}^{(1)}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{33g_2^2\lambda_{HS_2}^{(1)}}{2}$	$+3g_1^4$	$-\frac{15g_1^2\lambda_{HS_2}^{(1)}}{2}$

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 3 of 4

Column	Representative term structure
T1	$+\lambda_{HS_1}^{(1)}(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$
T2	$+6\lambda_1\lambda_{HS_2}^{(1)}$
T3	$+4\lambda_{HS_2}^{(1)^2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\lambda_{HS_1}^{(1)}(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)^2}$

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 4 of 4

Column	Representative term structure
T1	$+8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$
T2	$+6\lambda_{HS_2}^{(A)^2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	$+6\lambda_{HS_2}^{(A)^2}$

$\beta_{\lambda_{12}^{(1)}}^{(1)}$

Quantity: $\lambda_{12}^{(1)}$

$16\pi^2\,\mu\,\frac{d\lambda_{12}^{(1)}}{d\mu}=\beta_{\lambda_{12}^{(1)}}^{(1)}$

Terms 1 of 5

Column	Representative term structure
T1	$\frac{4}{3}\lambda_5\left(\lambda_5\right)^*$
T2	$+16g_2^4$
T3	$+g_2^2(-24\lambda_{12}^{(1)}-16\lambda_{12}^{(\times)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{4}{3}\lambda_5\left(\lambda_5\right)^*$	$+16g_2^4$	$+g_2^2(-24\lambda_{12}^{(1)}-16\lambda_{12}^{(\times)})$

Quantity: $\lambda_{12}^{(1)}$

Terms 2 of 5

Column	Representative term structure
T1	$+g_1^2(-6\lambda_{12}^{(1)}-4\lambda_{12}^{(\times)})$
T2	$+4\lambda_{12}^{(1)2}$
T3	$+\lambda_{S_1}^{(1)}\left(8\lambda_{12}^{(1)}+\frac{16\lambda_{12}^{(\times)}}{3}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+g_1^2(-6\lambda_{12}^{(1)}-4\lambda_{12}^{(\times)})$	$+4\lambda_{12}^{(1)2}$	$+\lambda_{S_1}^{(1)}\left(8\lambda_{12}^{(1)}+\frac{16\lambda_{12}^{(\times)}}{3}\right)$

Quantity: $\lambda_{12}^{(1)}$

Terms 3 of 5

Column	Representative term structure
T1	$+ \lambda_{12}^{(1)} \left(\frac{16 \lambda_{12}^{(\times)}}{3} + 4 \lambda_{S_1}^{(A)} + 4 \lambda_{S_2}^{(A)} \right)$
T2	$+ \lambda_{S_2}^{(1)} \left(8 \lambda_{12}^{(1)} + \frac{16 \lambda_{12}^{(\times)}}{3} \right)$
T3	$+ 16 \lambda_{12}^{(A)2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+ \lambda_{12}^{(1)} \left(\frac{16 \lambda_{12}^{(\times)}}{3} + 4 \lambda_{S_1}^{(A)} + 4 \lambda_{S_2}^{(A)} \right)$	$+ \lambda_{S_2}^{(1)} \left(8 \lambda_{12}^{(1)} + \frac{16 \lambda_{12}^{(\times)}}{3} \right)$	$+ 16 \lambda_{12}^{(A)2}$

Quantity: $\lambda_{12}^{(1)}$

Terms 4 of 5

Column	Representative term structure
T1	$+\frac{32\lambda_{12}^{(\times)2}}{3}$
T2	$+\frac{8\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}}{3}$
T3	$+\frac{8\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}}{3}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{32\lambda_{12}^{(\times)2}}{3}$	$+\frac{8\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}}{3}$	$+\frac{8\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}}{3}$

Quantity: $\lambda_{12}^{(1)}$

Terms 5 of 5

Column	Representative term structure
T1	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$

$\beta_{\lambda_5}^{(1)}$
Quantity: λ_5

$$16\pi^2\,\mu\,\frac{d\lambda_5}{d\mu}=\beta_{\lambda_5}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\lambda_5\left(6\,\text{Tr}\left(Y_d\,\left(Y_d\right)^\dagger\right)+2\,\text{Tr}\left(Y_e\,\left(Y_e\right)^\dagger\right)+6\,\text{Tr}\left(Y_u\,\left(Y_u\right)^\dagger\right)-\frac{4\lambda_{12}^{(A)}}{3}-2\lambda_{12}^{(\times)}-2\lambda_{HS_1}^{(A)}-\frac{4\lambda_{HS_2}^{(A)}}{3}\right)$
T2	$-\frac{33g_2^2\lambda_5}{2}$
T3	$-\frac{9g_1^2\lambda_5}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_5\left(6\,\text{Tr}\left(Y_d\,\left(Y_d\right)^\dagger\right)+2\,\text{Tr}\left(Y_e\,\left(Y_e\right)^\dagger\right)+6\,\text{Tr}\left(Y_u\,\left(Y_u\right)^\dagger\right)-\frac{4\lambda_{12}^{(A)}}{3}-2\lambda_{12}^{(\times)}-2\lambda_{HS_1}^{(A)}-\frac{4\lambda_{HS_2}^{(A)}}{3}\right)$	$-\frac{33g_2^2\lambda_5}{2}$	$-\frac{9g_1^2\lambda_5}{2}$

Quantity: λ_5

Terms 2 of 3

Column	Representative term structure
T1	$+2\lambda_{12}^{(1)}\lambda_5$
T2	$+2\lambda_1\lambda_5$
T3	$+4\lambda_{HS_1}^{(1)}\lambda_5$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$

Quantity: λ_5

Terms 3 of 3

Column	Representative term structure
T1	$+4\lambda_{HS_2}^{(1)}\lambda_5$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{HS_2}^{(1)}\lambda_5$

$\beta_{\lambda_{12}^{(A)}}^{(1)}$

Quantity: $\lambda_{12}^{(A)}$

$$16\pi^2\mu\frac{d\lambda_{12}^{(A)}}{d\mu}=\beta_{\lambda_{12}^{(A)}}^{(1)}$$

Terms 1 of 4

Column	Representative term structure
T1	$-\frac{2}{9}\lambda_5\left(\lambda_5\right)^*$
T2	$-24g_2{}^2\lambda_{12}^{(A)}$
T3	$-6g_1{}^2\lambda_{12}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{2}{9}\lambda_5\left(\lambda_5\right)^*$	$-24g_2{}^2\lambda_{12}^{(A)}$	$-6g_1{}^2\lambda_{12}^{(A)}$

Quantity: $\lambda_{12}^{(A)}$

Terms 2 of 4

Column	Representative term structure
T1	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$
T2	$+\frac{16\lambda_{12}^{(A)}\lambda_{12}^{(\times)}}{3}$
T3	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$	$+\frac{16\lambda_{12}^{(A)}\lambda_{12}^{(\times)}}{3}$	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$

Quantity: $\lambda_{12}^{(A)}$

Terms 3 of 4

Column	Representative term structure
T1	$+16\lambda_{12}^{(A)}\lambda_{S_1}^{(A)}$
T2	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$
T3	$+16\lambda_{12}^{(A)}\lambda_{S_2}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+16\lambda_{12}^{(A)}\lambda_{S_1}^{(A)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	$+16\lambda_{12}^{(A)}\lambda_{S_2}^{(A)}$

Quantity: $\lambda_{12}^{(A)}$

Terms 4 of 4

Column	Representative term structure
T1	$+3\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+3\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$

$\beta_{\lambda_{12}^{(\times)}}^{(1)}$

Quantity: $\lambda_{12}^{(\times)}$

$$16\pi^2\,\mu\,\frac{d\lambda_{12}^{(\times)}}{d\mu}=\beta_{\lambda_{12}^{(\times)}}^{(1)}$$

Terms 1 of 5

Column	Representative term structure
T1	$-\frac{1}{2}\lambda_5\,(\lambda_5)^*$
T2	$+\frac{33g_2^4}{4}$
T3	$+g_2^2(-6\lambda_{12}^{(1)}-24\lambda_{12}^{(\times)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{1}{2}\lambda_5\,(\lambda_5)^*$	$+\frac{33g_2^4}{4}$	$+g_2^2(-6\lambda_{12}^{(1)}-24\lambda_{12}^{(\times)})$

Quantity: $\lambda_{12}^{(\times)}$

Terms 2 of 5

Column	Representative term structure
T1	$+g_1^2 \left(-\frac{3\lambda_{12}^{(1)}}{2} - 6\lambda_{12}^{(\times)} \right)$
T2	$+\lambda_{12}^{(1)2}$
T3	$+\lambda_{S_1}^{(1)}(2\lambda_{12}^{(1)} + 3\lambda_{12}^{(\times)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+g_1^2 \left(-\frac{3\lambda_{12}^{(1)}}{2} - 6\lambda_{12}^{(\times)} \right)$	$+\lambda_{12}^{(1)2}$	$+\lambda_{S_1}^{(1)}(2\lambda_{12}^{(1)} + 3\lambda_{12}^{(\times)})$

Quantity: $\lambda_{12}^{(\times)}$

Terms 3 of 5

Column	Representative term structure
T1	$+\lambda_{12}^{(1)}(8\lambda_{12}^{(\times)}+\lambda_{S_1}^{(A)}+\lambda_{S_2}^{(A)})$
T2	$+\lambda_{S_2}^{(1)}(2\lambda_{12}^{(1)}+3\lambda_{12}^{(\times)})$
T3	$+4\lambda_{12}^{(A)2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\lambda_{12}^{(1)}(8\lambda_{12}^{(\times)}+\lambda_{S_1}^{(A)}+\lambda_{S_2}^{(A)})$	$+\lambda_{S_2}^{(1)}(2\lambda_{12}^{(1)}+3\lambda_{12}^{(\times)})$	$+4\lambda_{12}^{(A)2}$

Quantity: $\lambda_{12}^{(\times)}$

Terms 4 of 5

Column	Representative term structure
T1	$+11\lambda_{12}^{(\times)2}$
T2	$-\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$
T3	$-\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+11\lambda_{12}^{(\times)2}$	$-\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$	$-\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$

Quantity: $\lambda_{12}^{(\times)}$

Terms 5 of 5

Column	Representative term structure
T1	$+\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$

$$\beta_{\lambda_{HS_1}^{(A)}}^{(1)}$$

Quantity: $\lambda_{HS_1}^{(A)}$

$$16\pi^2\mu\frac{d\lambda_{HS_1}^{(A)}}{d\mu}=\beta_{\lambda_{HS_1}^{(A)}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\lambda_{HS_1}^{(A)}\left(6\,\mathrm{Tr}\Big(Y_d\,\,(Y_d)^\dagger\Big)+2\,\mathrm{Tr}\Big(Y_e\,\,(Y_e)^\dagger\Big)+6\,\mathrm{Tr}\Big(Y_u\,\,(Y_u)^\dagger\Big)+16\lambda_{S_1}^{(A)}\right)$
T2	$-\frac{4}{3}\lambda_5\,(\lambda_5)^*$
T3	$-\frac{33g_2^2\lambda_{HS_1}^{(A)}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_{HS_1}^{(A)}\left(6\,\mathrm{Tr}\Big(Y_d\,\,(Y_d)^\dagger\Big)+2\,\mathrm{Tr}\Big(Y_e\,\,(Y_e)^\dagger\Big)+6\,\mathrm{Tr}\Big(Y_u\,\,(Y_u)^\dagger\Big)+16\lambda_{S_1}^{(A)}\right)$	$-\frac{4}{3}\lambda_5\,(\lambda_5)^*$	$-\frac{33g_2^2\lambda_{HS_1}^{(A)}}{2}$

Quantity: $\lambda_{HS_1}^{(A)}$

Terms 2 of 3

Column	Representative term structure
T1	$-\frac{3g_1^2\lambda_{HS_1}^{(A)}}{2}$
T2	$+12\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$
T3	$+2\lambda_1\lambda_{HS_1}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{3g_1^2\lambda_{HS_1}^{(A)}}{2}$	$+12\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$	$+2\lambda_1\lambda_{HS_1}^{(A)}$

Quantity: $\lambda_{HS_1}^{(A)}$

Terms 3 of 3

Column	Representative term structure
T1	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$
T2	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$

$$\beta_{\lambda_{HS_2}^{(A)}}^{(1)}$$

Quantity: $\lambda_{HS_2}^{(A)}$

$$16\pi^2\mu\frac{d\lambda_{HS_2}^{(A)}}{d\mu}=\beta_{\lambda_{HS_2}^{(A)}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\lambda_{HS_2}^{(A)}\left(6\,\mathrm{Tr}\left(Y_d\, \left(Y_d\right)^\dagger\right)+2\,\mathrm{Tr}\left(Y_e\, \left(Y_e\right)^\dagger\right)+6\,\mathrm{Tr}\left(Y_u\, \left(Y_u\right)^\dagger\right)+16\lambda_{S_2}^{(A)}\right)$
T2	$-\frac{8}{9}\lambda_5\left(\lambda_5\right)^*$
T3	$+g_1^2\left(4g_2^2-\frac{15\lambda_{HS_2}^{(A)}}{2}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_{HS_2}^{(A)}\left(6\,\mathrm{Tr}\left(Y_d\, \left(Y_d\right)^\dagger\right)+2\,\mathrm{Tr}\left(Y_e\, \left(Y_e\right)^\dagger\right)+6\,\mathrm{Tr}\left(Y_u\, \left(Y_u\right)^\dagger\right)+16\lambda_{S_2}^{(A)}\right)$	$-\frac{8}{9}\lambda_5\left(\lambda_5\right)^*$	$+g_1^2\left(4g_2^2-\frac{15\lambda_{HS_2}^{(A)}}{2}\right)$

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 2 of 3

Column	Representative term structure
T1	$-\frac{33g_2^2\lambda_{HS_2}^{(A)}}{2}$
T2	$+12\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$
T3	$+2\lambda_1\lambda_{HS_2}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{33g_2^2\lambda_{HS_2}^{(A)}}{2}$	$+12\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$	$+2\lambda_1\lambda_{HS_2}^{(A)}$

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 3 of 3

Column	Representative term structure
T1	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$
T2	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$

$\beta_{\lambda_{S_1}^{(A)}}^{(1)}$

Quantity: $\lambda_{S_1}^{(A)}$

$16\pi^2\mu\frac{d\lambda_{S_1}^{(A)}}{d\mu}=\beta_{\lambda_{S_1}^{(A)}}^{(1)}$

Terms 1 of 4

Column	Representative term structure
T1	$\frac{3g_2^4}{4}$
T2	$+g_2^2(-3\lambda_{S_1}^{(1)}-24\lambda_{S_1}^{(A)})$
T3	$+\frac{3\lambda_{12}^{(1)2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{3g_2^4}{4}$	$+g_2^2(-3\lambda_{S_1}^{(1)}-24\lambda_{S_1}^{(A)})$	$+\frac{3\lambda_{12}^{(1)2}}{4}$

Quantity: $\lambda_{S_1}^{(A)}$

Terms 2 of 4

Column	Representative term structure
T1	$+\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$
T2	$+12\lambda_{12}^{(A)2}$
T3	$-\frac{\lambda_{12}^{(\times)2}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	$+12\lambda_{12}^{(A)2}$	$-\frac{\lambda_{12}^{(\times)2}}{2}$

Quantity: $\lambda_{S_1}^{(A)}$

Terms 3 of 4

Column	Representative term structure
T1	$+\frac{\lambda_{HS_1}^{(1) \ 2}}{2}$
T2	$+3\lambda_{HS_1}^{(A) \ 2}$
T3	$+\frac{7\lambda_{S_1}^{(1) 2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{\lambda_{HS_1}^{(1) \ 2}}{2}$	$+3\lambda_{HS_1}^{(A) \ 2}$	$+\frac{7\lambda_{S_1}^{(1) 2}}{4}$

Quantity: $\lambda_{S_1}^{(A)}$

Terms 4 of 4

Column	Representative term structure
T1	$+13\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$
T2	$+22\lambda_{S_1}^{(A)2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+13\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$	$+22\lambda_{S_1}^{(A)2}$

$\beta_{\lambda_{S_2}}^{(1)(A)}$

Quantity: $\lambda_{S_2}^{(A)}$

$16\pi^2\mu\frac{d\lambda_{S_2}^{(A)}}{d\mu}=\beta_{\lambda_{S_2}}^{(1)(A)}$

Terms 1 of 5

Column	Representative term structure						
T1	$\frac{3g_2^4}{4}$						
T2	$+g_1^2\left(9g_2^2-\frac{3\lambda_{S_2}^{(1)}}{2}-12\lambda_{S_2}^{(A)}\right)$						
T3	$+g_2^2(-3\lambda_{S_2}^{(1)}-24\lambda_{S_2}^{(A)})$						

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{3g_2^4}{4}$	$+g_1^2\left(9g_2^2-\frac{3\lambda_{S_2}^{(1)}}{2}-12\lambda_{S_2}^{(A)}\right)$	$+g_2^2(-3\lambda_{S_2}^{(1)}-24\lambda_{S_2}^{(A)})$

Quantity: $\lambda_{S_2}^{(A)}$

Terms 2 of 5

Column	Representative term structure
T1	$+ \frac{3g_1^4}{2}$
T2	$+ \frac{3\lambda_{12}^{(1)2}}{4}$
T3	$+ \lambda_{12}^{(1)} \lambda_{12}^{(\times)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+ \frac{3g_1^4}{2}$	$+ \frac{3\lambda_{12}^{(1)2}}{4}$	$+ \lambda_{12}^{(1)} \lambda_{12}^{(\times)}$

Quantity: $\lambda_{S_2}^{(A)}$

Terms 3 of 5

Column	Representative term structure
T1	$+12\lambda_{12}^{(A)2}$
T2	$-\frac{\lambda_{12}^{(\times)2}}{2}$
T3	$+\frac{\lambda_{HS_2}^{(1)2}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+12\lambda_{12}^{(A)2}$	$-\frac{\lambda_{12}^{(\times)2}}{2}$	$+\frac{\lambda_{HS_2}^{(1)2}}{2}$

Quantity: $\lambda_{S_2}^{(A)}$

Terms 4 of 5

Column	Representative term structure
T1	$+3\lambda_{HS_2}^{(A) \ 2}$
T2	$+\frac{7\lambda_{S_2}^{(1) 2}}{4}$
T3	$+13\lambda_{S_2}^{(1)} \lambda_{S_2}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+3\lambda_{HS_2}^{(A) \ 2}$	$+\frac{7\lambda_{S_2}^{(1) 2}}{4}$	$+13\lambda_{S_2}^{(1)} \lambda_{S_2}^{(A)}$

Quantity: $\lambda_{S_2}^{(A)}$

Terms 5 of 5

Column	Representative term structure
T1	$+22\lambda_{S_2}^{(A)^2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+22\lambda_{S_2}^{(A)^2}$

Dimension-five $\psi^2\phi^2$ Wilson coefficients

The Wilson tensor is reported in the real-scalar component basis C_{ijab} . Each component with a nonzero one-loop beta function has its own table. Components absent at the matching boundary but generated by operator mixing are included automatically.

$$\beta_{C_{1133}}^{(1)}$$

Quantity: 1, 1, 3, 3

$$16\pi^2\mu\frac{dC_{1133}}{d\mu}=\beta_{C_{1133}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{6}lambdaT_3\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{2\sqrt{6}lambdaT_3\overline{y_1y_2}}{3MF}$

$\beta_{C_{1134}}^{(1)}$

Quantity: 1, 1, 3, 4

$$16\pi^2\,_{\mu}\frac{dC_{1134}}{d\mu}=\beta_{C_{1134}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{6}i\lambda bdaT_3y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{2\sqrt{6}i\lambda bdaT_3y_1y_2}{3MF}$

$\beta_{C_{1144}}^{(1)}$

Quantity: 1, 1, 4, 4

$$16\pi^2\,_{\mu}\frac{dC_{1144}}{d\mu}=\beta_{C_{1144}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{6}lambdaT_3\overline{y_1}y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{2\sqrt{6}lambdaT_3\overline{y_1}y_2}{3MF}$

$$16\pi^2\,\mu\frac{dC_{11513}}{d\mu}=\beta_{C_{11513}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{11}\lambda da12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{11\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}\lambda da12Inv2\overline{y_1y_2}}{22MF}$	$-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$

Quantity: 1, 1, 5, 13

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}lambda12Inv1y_1y_2}{MF}$
T2	$\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}lambda12Inv1y_1y_2}{6MF}$	$-\frac{\sqrt{3}ig_2^2y_1y_2}{6MF}$	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$

$$16\pi^2\,\mu\frac{dC_{11514}}{d\mu}=\beta_{C_{11514}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{11}i\lambda da12Inv2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{11\sqrt{3}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}i\lambda da12Inv2\overline{y_1y_2}}{22MF}$

Quantity: 1, 1, 5, 14

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}iye\overline{y_1y_2ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}iye\overline{y_1y_2ye}}{12MF}$

$$16\pi^2\,\mu\frac{dC_{11613}}{d\mu}=\beta_{C_{11613}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$

Quantity: 1, 1, 6, 13

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{11} i \lambda d a 12 I n v 2 y_1 y_2}{M F}$
T2	$\frac{\sqrt{3} i g_2^2 y_1 y_2}{M F}$
T3	$\frac{\sqrt{3} i y e \overline{y_1 y_2 y e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{3 \sqrt{11} i \lambda d a 12 I n v 2 y_1 y_2}{22 M F}$	$\frac{11 \sqrt{3} i g_2^2 y_1 y_2}{8 M F}$	$- \frac{\sqrt{3} i y e \overline{y_1 y_2 y e}}{12 M F}$

$$16\pi^2\,\mu\frac{dC_{11614}}{d\mu}=\beta_{C_{11614}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{11}\lambda da12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{11\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}\lambda da12Inv2\overline{y_1y_2}}{22MF}$	$-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$

Quantity: 1, 1, 6, 14

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}lambda12Inv1y_1y_2}{MF}$
T2	$\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}lambda12Inv1y_1y_2}{6MF}$	$-\frac{\sqrt{3}ig_2^2y_1y_2}{6MF}$	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$

$\beta_{C_{11711}}^{(1)}$

Quantity: 1, 1, 7, 11

$$16\pi^2\,\mu\frac{dC_{11711}}{d\mu}=\beta_{C_{11711}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{11}lambda12Inv2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$

Quantity: 1, 1, 7, 11

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1}y_2}{MF}$
T2	$\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{MF}$
T3	$\frac{\sqrt{6}ig_2^2\overline{y_1}y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{49\sqrt{3}g_2^2\overline{y_1}y_2}{24MF}$	$-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{12MF}$	$\frac{\sqrt{6}ig_2^2\overline{y_1}y_2}{3MF}$

Quantity: 1, 1, 7, 11

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}ig_2^2y_1y_2}{6MF}$

$$16\pi^2\,\mu\frac{dC_{11712}}{d\mu}=\beta_{C_{11712}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$

Quantity: 1, 1, 7, 12

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$	$\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$	$\frac{49\sqrt{3}ig_2^2\overline{y_1y_2}}{24MF}$

Quantity: 1, 1, 7, 12

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3} i y e y_1 y_2 y \bar{e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3} i y e y_1 y_2 y \bar{e}}{12 M F}$

$$_{16\pi^2}\mu\frac{dC_{11715}}{d\mu}=\beta_{C_{11715}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$

Quantity: 1, 1, 7, 15

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{11} i \lambda d a 12 I n v 2 y_1 y_2}{M F}$
T2	$\frac{\sqrt{3} i g_2^2 y_1 y_2}{M F}$
T3	$\frac{\sqrt{3} i y e \overline{y_1 y_2 y e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{3 \sqrt{11} i \lambda d a 12 I n v 2 y_1 y_2}{22 M F}$	$\frac{11 \sqrt{3} i g_2^2 y_1 y_2}{8 M F}$	$- \frac{\sqrt{3} i y e \overline{y_1 y_2 y e}}{12 M F}$

$$_{16\pi^2}^2\mu\frac{dC_{11716}}{d\mu}=\beta_{C_{11716}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{11}\lambda da12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{11\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}\lambda da12Inv2\overline{y_1y_2}}{22MF}$	$-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$

Quantity: 1, 1, 7, 16

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$

$$16\pi^2\,\mu\frac{dC_{11811}}{d\mu}=\beta_{C_{11811}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{6}g_2^2y_1y_2}{MF}$
T2	$\frac{\sqrt{3}g_2^2y_1y_2}{MF}$
T3	$\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$	$\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$	$-\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$

Quantity: 1, 1, 8, 11

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{11}ilambda12Inv2\overline{y_1}y_2}{MF}$
T2	$\frac{\sqrt{3}ig_1^2\overline{y_1}y_2}{MF}$
T3	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1}y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{3\sqrt{11}ilambda12Inv2\overline{y_1}y_2}{22MF}$	$-\frac{\sqrt{3}ig_1^2\overline{y_1}y_2}{8MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1}y_2}{6MF}$

Quantity: 1, 1, 8, 11

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3} i y e y_1 y_2 y \bar{e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3} i y e y_1 y_2 y \bar{e}}{12 M F}$

$$16\pi^2\,\mu\frac{dC_{11812}}{d\mu}=\beta_{C_{11812}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{11}lambda12Inv2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$

Quantity: 1, 1, 8, 12

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1}y_2}{MF}$
T2	$\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{MF}$
T3	$\frac{\sqrt{6}ig_2^2\overline{y_1}y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{49\sqrt{3}g_2^2\overline{y_1}y_2}{24MF}$	$-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{12MF}$	$\frac{\sqrt{6}ig_2^2\overline{y_1}y_2}{3MF}$

Quantity: 1, 1, 8, 12

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}ig_2^2y_1y_2}{6MF}$

$$16\pi^2\,\mu\frac{dC_{11815}}{d\mu}=\beta_{C_{11815}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{11}lambda12Inv2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1}y_2}{MF}$
T2	$\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{11\sqrt{3}g_2^2\overline{y_1}y_2}{8MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{6MF}$	$-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{12MF}$

$$_{16\pi^2}^2\mu\frac{dC_{11816}}{d\mu}=\beta_{C_{11816}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$

Quantity: 1, 1, 8, 16

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{11} i \lambda d a 12 I n v 2 y_1 y_2}{M F}$
T2	$\frac{\sqrt{3} i g_2^2 y_1 y_2}{M F}$
T3	$\frac{\sqrt{3} i y e \overline{y_1 y_2 y e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{3 \sqrt{11} i \lambda d a 12 I n v 2 y_1 y_2}{22 M F}$	$\frac{11 \sqrt{3} i g_2^2 y_1 y_2}{8 M F}$	$- \frac{\sqrt{3} i y e \overline{y_1 y_2 y e}}{12 M F}$

$$16\pi^2\,\mu\frac{dC_{11913}}{d\mu}=\beta_{C_{11913}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{11}i\lambda da12Inv2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{49\sqrt{3}ig_2^2\overline{y_1y_2}}{24MF}$	$-\frac{3\sqrt{11}i\lambda da12Inv2\overline{y_1y_2}}{22MF}$

Quantity: 1, 1, 9, 13

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ilambda12In\nu1\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$	$-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$	$\frac{\sqrt{3}ilambda12In\nu1\overline{y_1y_2}}{6MF}$

Quantity: 1, 1, 9, 13

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3} i y e y_1 y_2 y \bar{e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3} i y e y_1 y_2 y \bar{e}}{12 M F}$

$$_{16\pi^2}\mu\frac{dC_{11914}}{d\mu}=\beta_{C_{11914}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$

Quantity: 1, 1, 9, 14

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{11}\lambda a_{12} \operatorname{Inv} 2 \overline{y_1 y_2}}{MF}$
T2	$\frac{\sqrt{3} g_2^2 \overline{y_1 y_2}}{MF}$
T3	$\frac{\sqrt{3} i g_2^2 \overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{3\sqrt{11}\lambda a_{12} \operatorname{Inv} 2 \overline{y_1 y_2}}{22MF}$	$\frac{49\sqrt{3} g_2^2 \overline{y_1 y_2}}{24MF}$	$-\frac{\sqrt{3} i g_2^2 \overline{y_1 y_2}}{6MF}$

Quantity: 1, 1, 9, 14

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}ye\overline{y}_1y_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}ye\overline{y}_1y_2\overline{y}e}{12MF}$

$$16\pi^2\mu\frac{dC_{1111013}}{d\mu}=\beta_{C_{1111013}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{11}\lambda da12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{49\sqrt{3}g_2^2\overline{y_1y_2}}{24MF}$	$-\frac{3\sqrt{11}\lambda da12Inv2\overline{y_1y_2}}{22MF}$	$-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$

Quantity: 1, 1, 10, 13

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}\lambda_{12}Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{\sqrt{3}\lambda_{12}Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$

Quantity: 1, 1, 10, 13

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}ye\overline{y}_1y_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}ye\overline{y}_1y_2\overline{y}e}{12MF}$

$$16\pi^2{}_{\mu}\frac{dC_{111014}}{d\mu}=\beta_{C_{111014}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{11}ilambda12In v2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{49\sqrt{3}ig_2^2\overline{y_1y_2}}{24MF}$	$-\frac{3\sqrt{11}ilambda12In v2\overline{y_1y_2}}{22MF}$

Quantity: 1, 1, 10, 14

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ilambda12In\nu1\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$	$-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$	$\frac{\sqrt{3}ilambda12In\nu1\overline{y_1y_2}}{6MF}$

Quantity: 1, 1, 10, 14

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3} i y e y_1 y_2 y \bar{e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3} i y e y_1 y_2 y \bar{e}}{12 M F}$

$$\beta_{C_{1213}}^{(1)}$$

Quantity: 1, 2, 1, 3

$$16\pi^2\,_{\mu}\frac{dC_{1213}}{d\mu}=\beta_{C_{1213}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{6}lambda T_3 \overline{y_1 y_2}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{6}lambda T_3 \overline{y_1 y_2}}{3M F}$

$\beta_{C_{1214}}^{(1)}$

Quantity: 1, 2, 1, 4

$$16\pi^2\,\mu\,\frac{dC_{1214}}{d\mu}=\beta_{C_{1214}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{6}i\lambda bdaT_3\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{6}i\lambda bdaT_3\overline{y_1y_2}}{3MF}$

$\beta_{C_{1223}}^{(1)}$

Quantity: 1, 2, 2, 3

$$16\pi^2\,_{\mu}\frac{dC_{1223}}{d\mu}=\beta_{C_{1223}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{6}i\lambda b d a T_3 \overline{y_1 y_2}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{6}i\lambda b d a T_3 \overline{y_1 y_2}}{3 M F}$

$\beta_{C_{1224}}^{(1)}$

Quantity: 1, 2, 2, 4

$$16\pi^2\,\mu\frac{dC_{1224}}{d\mu}=\beta_{C_{1224}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{6}lambdaT_3\overline{y_1}y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{6}lambdaT_3\overline{y_1}y_2}{3MF}$

$\beta_{C_{12511}}^{(1)}$
Quantity: 1, 2, 5, 11

$$16\pi^2\,\mu\frac{dC_{12511}}{d\mu}=\beta_{C_{12511}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{6}g_2^2y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$

$\beta_{C_{12512}}^{(1)}$

Quantity: 1, 2, 5, 12

$$16\pi^2\,\mu\frac{dC_{12512}}{d\mu}=\beta_{C_{12512}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{6}ig_2^2y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{6}ig_2^2y_1y_2}{3MF}$

$$16\pi^2\,\mu\frac{dC_{12515}}{d\mu}=\beta_{C_{12515}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$

Quantity: 1, 2, 5, 15

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$	$\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$	$\frac{15\sqrt{3}ig_2^2\overline{y_1y_2}}{8MF}$

Quantity: 1, 2, 5, 15

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3} i y e y_1 y_2 y \bar{e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3} i y e y_1 y_2 y \bar{e}}{12 M F}$

$$16\pi^2\,\mu\frac{dC_{12516}}{d\mu}=\beta_{C_{12516}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{11}\lambda da12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{15\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}\lambda da12Inv2\overline{y_1y_2}}{22MF}$	$-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$

Quantity: 1, 2, 5, 16

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$

Quantity: 1, 2, 5, 16

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}ye\overline{y}_1y_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}ye\overline{y}_1y_2\overline{y}e}{12MF}$

$\beta_{C_{12611}}^{(1)}$

Quantity: 1, 2, 6, 11

$$16\pi^2\,\mu\frac{dC_{12611}}{d\mu}=\beta_{C_{12611}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{6}ig_2^2y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{6}ig_2^2y_1y_2}{3MF}$

$\beta_{C_{12612}}^{(1)}$

Quantity: 1, 2, 6, 12

$$16\pi^2\,\mu\frac{dC_{12612}}{d\mu}=\beta_{C_{12612}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{6}g_2^2y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$

$$16\pi^2\,\mu\frac{dC_{12615}}{d\mu}=\beta_{C_{12615}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$

Quantity: 1, 2, 6, 15

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{11}\lambda a_{12} \text{Inv} 2 \overline{y_1 y_2}}{MF}$
T2	$\frac{\sqrt{3} g_2^2 \overline{y_1 y_2}}{MF}$
T3	$\frac{\sqrt{3} y e \overline{y_1 y_2 y e}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{3\sqrt{11}\lambda a_{12} \text{Inv} 2 \overline{y_1 y_2}}{22MF}$	$\frac{15\sqrt{3} g_2^2 \overline{y_1 y_2}}{8MF}$	$-\frac{\sqrt{3} y e \overline{y_1 y_2 y e}}{12MF}$

Quantity: 1, 2, 6, 15

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{6}ig_2^2y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{6}ig_2^2y_1y_2}{6MF}$

$$_{16\pi^2}^2\mu\frac{dC_{12616}}{d\mu}=\beta_{C_{12616}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}i\lambda da12Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}i\lambda da12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$

Quantity: 1, 2, 6, 16

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$	$\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$	$\frac{15\sqrt{3}ig_2^2\overline{y_1y_2}}{8MF}$

Quantity: 1, 2, 6, 16

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3} i y e y_1 y_2 y \bar{e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3} i y e y_1 y_2 y \bar{e}}{12 M F}$

$\beta_{C_{12713}}^{(1)}$

Quantity: 1, 2, 7, 13

$$16\pi^2\,\mu\frac{dC_{12713}}{d\mu}=\beta_{C_{12713}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{6}g_2^2y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$

$\beta_{C_{12714}}^{(1)}$

Quantity: 1, 2, 7, 14

$$16\pi^2\,\mu\frac{dC_{12714}}{d\mu}=\beta_{C_{12714}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{6}ig_2^2y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{6}ig_2^2y_1y_2}{3MF}$

$\beta_{C_{12813}}^{(1)}$

Quantity: 1, 2, 8, 13

$$16\pi^2\,\mu\frac{dC_{12813}}{d\mu}=\beta_{C_{12813}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{6}ig_2^2y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{6}ig_2^2y_1y_2}{3MF}$

$\beta_{C_{12814}}^{(1)}$
Quantity: 1, 2, 8, 14

$$16\pi^2\,\mu\frac{dC_{12814}}{d\mu}=\beta_{C_{12814}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{6}g_2^2y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$

$$16\pi^2\,\mu\frac{dC_{12911}}{d\mu}=\beta_{C_{12911}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{11}i\lambda da12Inv2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{15\sqrt{3}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}i\lambda da12Inv2\overline{y_1y_2}}{22MF}$

Quantity: 1, 2, 9, 11

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ilambda12In\nu1\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$	$\frac{\sqrt{3}ilambda12In\nu1\overline{y_1y_2}}{6MF}$

Quantity: 1, 2, 9, 11

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3} i y e y_1 y_2 y \bar{e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3} i y e y_1 y_2 y \bar{e}}{12 M F}$

$$16\pi^2\,\mu\frac{dC_{12912}}{d\mu}=\beta_{C_{12912}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$

Quantity: 1, 2, 9, 12

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{11}\lambda a_{12} \operatorname{Inv} 2 \overline{y_1 y_2}}{MF}$
T2	$\frac{\sqrt{3} g_2^2 \overline{y_1 y_2}}{MF}$
T3	$\frac{\sqrt{6} i g_2^2 \overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{3\sqrt{11}\lambda a_{12} \operatorname{Inv} 2 \overline{y_1 y_2}}{22MF}$	$\frac{15\sqrt{3} g_2^2 \overline{y_1 y_2}}{8MF}$	$-\frac{\sqrt{6} i g_2^2 \overline{y_1 y_2}}{6MF}$

Quantity: 1, 2, 9, 12

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}ye\overline{y}_1y_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}ye\overline{y}_1y_2\overline{y}e}{12MF}$

$$16\pi^2\mu\frac{dC_{121011}}{d\mu}=\beta_{C_{121011}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{11}\overline{\lambda}da12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{15\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}\overline{\lambda}da12Inv2\overline{y_1y_2}}{22MF}$	$-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$

Quantity: 1, 2, 10, 11

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}\lambda_{12}Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{\sqrt{3}\lambda_{12}Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$

Quantity: 1, 2, 10, 11

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}ye\overline{y}_1y_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}ye\overline{y}_1y_2\overline{y}e}{12MF}$

$$16\pi^2\mu\frac{dC_{121012}}{d\mu}=\beta_{C_{121012}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{11}i\lambda da12Inv2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{15\sqrt{3}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}i\lambda da12Inv2\overline{y_1y_2}}{22MF}$

Quantity: 1, 2, 10, 12

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ilambda12In\nu1\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$	$\frac{\sqrt{3}ilambda12In\nu1\overline{y_1y_2}}{6MF}$

Quantity: 1, 2, 10, 12

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3} i y e y_1 y_2 y \bar{e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3} i y e y_1 y_2 y \bar{e}}{12 M F}$

$\beta_{C_{2211}}^{(1)}$

Quantity: 2, 2, 1, 1

$$16\pi^2\,_{\mu}\frac{dC_{2211}}{d\mu}=\beta_{C_{2211}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{6}lambdaT_3\overline{y_1}y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{2\sqrt{6}lambdaT_3\overline{y_1}y_2}{3MF}$

$\beta_{C_{2212}}^{(1)}$

Quantity: 2, 2, 1, 2

$$16\pi^2\,_{\mu}\frac{dC_{2212}}{d\mu}=\beta_{C_{2212}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{6}i\lambda bdaT_3y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{2\sqrt{6}i\lambda bdaT_3y_1y_2}{3MF}$

$\beta_{C_{2222}}^{(1)}$

Quantity: 2, 2, 2, 2

$$16\pi^2\,_{\mu}\frac{dC_{2222}}{d\mu}=\beta_{C_{2222}}^{(1)}$$

Column	Representative term structure
T1	$\frac{\sqrt{6}lambdaT_3\overline{y_1}y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{2\sqrt{6}lambdaT_3\overline{y_1}y_2}{3MF}$

$$16\pi^2\,\mu\frac{dC_{22513}}{d\mu}=\beta_{C_{22513}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{11}\lambda da12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{49\sqrt{3}g_2^2\overline{y_1y_2}}{24MF}$	$-\frac{3\sqrt{11}\lambda da12Inv2\overline{y_1y_2}}{22MF}$	$-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$

Quantity: 2, 2, 5, 13

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$

Quantity: 2, 2, 5, 13

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}ye\overline{y}_1y_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}ye\overline{y}_1y_2\overline{y}e}{12MF}$

$$16\pi^2\,\mu\frac{dC_{22514}}{d\mu}=\beta_{C_{22514}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{6}g_2^2y_1y_2}{MF}$
T2	$\frac{\sqrt{3}g_2^2y_1y_2}{MF}$
T3	$\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{6}g_2^2y_1y_2}{3MF}$	$-\frac{\sqrt{3}g_2^2y_1y_2}{6MF}$	$-\frac{49\sqrt{3}ig_2^2y_1y_2}{24MF}$

Quantity: 2, 2, 5, 14

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{11}ilambda12Inv2\overline{y_1}y_2}{MF}$
T2	$\frac{\sqrt{3}ig_1^2\overline{y_1}y_2}{MF}$
T3	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1}y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{3\sqrt{11}ilambda12Inv2\overline{y_1}y_2}{22MF}$	$-\frac{\sqrt{3}ig_1^2\overline{y_1}y_2}{8MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1}y_2}{6MF}$

Quantity: 2, 2, 5, 14

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3} i y e y_1 y_2 y \bar{e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3} i y e y_1 y_2 y \bar{e}}{12 M F}$

$$16\pi^2\,\mu\frac{dC_{22613}}{d\mu}=\beta_{C_{22613}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$

Quantity: 2, 2, 6, 13

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$	$\frac{3\sqrt{11}ilambda12Inv2\overline{y_1y_2}}{22MF}$	$\frac{49\sqrt{3}ig_2^2\overline{y_1y_2}}{24MF}$

Quantity: 2, 2, 6, 13

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3} i y e y_1 y_2 y \bar{e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3} i y e y_1 y_2 y \bar{e}}{12 M F}$

$$16\pi^2\,\mu\frac{dC_{22614}}{d\mu}=\beta_{C_{22614}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{11}\overline{lambda12Inv2\overline{y_1y_2}}}{MF}$
T3	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{49\sqrt{3}g_2^2\overline{y_1y_2}}{24MF}$	$-\frac{3\sqrt{11}\overline{lambda12Inv2\overline{y_1y_2}}}{22MF}$	$-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$

Quantity: 2, 2, 6, 14

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}ye\overline{y}_1y_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}ye\overline{y}_1y_2\overline{y}e}{12MF}$

$\beta_{C_{22711}}^{(1)}$

Quantity: 2, 2, 7, 11

$$16\pi^2\,\mu\frac{dC_{22711}}{d\mu}=\beta_{C_{22711}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{11}lambda12Inv2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1}y_2}{MF}$
T2	$\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{11\sqrt{3}g_2^2\overline{y_1}y_2}{8MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{6MF}$	$-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{12MF}$

$$16\pi^2\,\mu\frac{dC_{22712}}{d\mu}=\beta_{C_{22712}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}i\lambda da12Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}i\lambda da12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{11} i \lambda b d a 12 I n v 2 y_1 y_2}{M F}$
T2	$\frac{\sqrt{3} i g_2^2 y_1 y_2}{M F}$
T3	$\frac{\sqrt{3} i y e \overline{y_1 y_2 y e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{3 \sqrt{11} i \lambda b d a 12 I n v 2 y_1 y_2}{22 M F}$	$\frac{11 \sqrt{3} i g_2^2 y_1 y_2}{8 M F}$	$- \frac{\sqrt{3} i y e \overline{y_1 y_2 y e}}{12 M F}$

$$16\pi^2\,\mu\frac{dC_{22715}}{d\mu}=\beta_{C_{22715}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{11}i\lambda da12Inv2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{49\sqrt{3}ig_2^2\overline{y_1y_2}}{24MF}$	$-\frac{3\sqrt{11}i\lambda da12Inv2\overline{y_1y_2}}{22MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ilambda12In\nu1\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$	$-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$	$\frac{\sqrt{3}ilambda12In\nu1\overline{y_1y_2}}{6MF}$

Quantity: 2, 2, 7, 15

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3} i y e y_1 y_2 y \bar{e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3} i y e y_1 y_2 y \bar{e}}{12 M F}$

$$_{16\pi^2}^2\mu\frac{dC_{22716}}{d\mu}=\beta_{C_{22716}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{11}\lambda a_{12} \text{Inv} 2 \overline{y_1 y_2}}{MF}$
T2	$\frac{\sqrt{3} g_2^2 \overline{y_1 y_2}}{MF}$
T3	$\frac{\sqrt{3} y e \overline{y_1 y_2 y e}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{3\sqrt{11}\lambda a_{12} \text{Inv} 2 \overline{y_1 y_2}}{22MF}$	$\frac{49\sqrt{3} g_2^2 \overline{y_1 y_2}}{24MF}$	$-\frac{\sqrt{3} y e \overline{y_1 y_2 y e}}{12MF}$

Quantity: 2, 2, 7, 16

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}ig_2^2y_1y_2}{6MF}$

$$16\pi^2\,\mu\frac{dC_{22811}}{d\mu}=\beta_{C_{22811}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{11}i\lambda a_{12}nv2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{11\sqrt{3}ig_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}i\lambda a_{12}nv2\overline{y_1y_2}}{22MF}$

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}iye\overline{y_1y_2}ye}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}iye\overline{y_1y_2}ye}{12MF}$

$\beta_{C_{22812}}^{(1)}$
Quantity: 2, 2, 8, 12

$$16\pi^2\,\mu\frac{dC_{22812}}{d\mu}=\beta_{C_{22812}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{11}lambda12Inv2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1}y_2}{MF}$
T2	$\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{11\sqrt{3}g_2^2\overline{y_1}y_2}{8MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{6MF}$	$-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y}e}{12MF}$

$$16\pi^2\,\mu\frac{dC_{22815}}{d\mu}=\beta_{C_{22815}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{11}\lambda da12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{49\sqrt{3}g_2^2\overline{y_1y_2}}{24MF}$	$-\frac{3\sqrt{11}\lambda da12Inv2\overline{y_1y_2}}{22MF}$	$-\frac{\sqrt{6}g_2^2\overline{y_1y_2}}{3MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}\lambda\text{Inv1}\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{\sqrt{3}\lambda\text{Inv1}\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{6MF}$

Quantity: 2, 2, 8, 15

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}ye\overline{y}_1y_2\overline{y}e}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}ye\overline{y}_1y_2\overline{y}e}{12MF}$

$$16\pi^2\,\mu\frac{dC_{22816}}{d\mu}=\beta_{C_{22816}}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_2^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{11}i\lambda da12Inv2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{49\sqrt{3}ig_2^2\overline{y_1y_2}}{24MF}$	$-\frac{3\sqrt{11}i\lambda da12Inv2\overline{y_1y_2}}{22MF}$

Terms 2 of 3

Column	Representative term structure
T1	$\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ilambda12In\nu1\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{6}ig_2^2\overline{y_1y_2}}{3MF}$	$-\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$	$\frac{\sqrt{3}ilambda12In\nu1\overline{y_1y_2}}{6MF}$

Terms 3 of 3

Column	Representative term structure
T1	$\frac{\sqrt{3} i y e y_1 y_2 y \bar{e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3} i y e y_1 y_2 y \bar{e}}{12 M F}$

$$16\pi^2\,\mu\frac{dC_{22913}}{d\mu}=\beta_{C_{22913}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}ilambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$

Quantity: 2, 2, 9, 13

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{11} i \lambda d a 12 I n v 2 y_1 y_2}{M F}$
T2	$\frac{\sqrt{3} i g_2^2 y_1 y_2}{M F}$
T3	$\frac{\sqrt{3} i y e \overline{y_1 y_2 y e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{3 \sqrt{11} i \lambda d a 12 I n v 2 y_1 y_2}{22 M F}$	$\frac{11 \sqrt{3} i g_2^2 y_1 y_2}{8 M F}$	$- \frac{\sqrt{3} i y e \overline{y_1 y_2 y e}}{12 M F}$

$$16\pi^2\,\mu\frac{dC_{22914}}{d\mu}=\beta_{C_{22914}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{11}\lambda da12Inv2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{11\sqrt{3}g_2^2\overline{y_1y_2}}{8MF}$	$-\frac{3\sqrt{11}\lambda da12Inv2\overline{y_1y_2}}{22MF}$	$-\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}lambda12Inv1y_1y_2}{MF}$
T2	$\frac{\sqrt{3}ig_2^2y_1y_2}{MF}$
T3	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3}lambda12Inv1y_1y_2}{6MF}$	$-\frac{\sqrt{3}ig_2^2y_1y_2}{6MF}$	$\frac{\sqrt{3}ye\overline{y_1y_2ye}}{12MF}$

$\beta_{C_{221013}}^{(1)}$
Quantity: 2, 2, 10, 13

$$16\pi^2{}_{\mu}\frac{dC_{221013}}{d\mu}=\beta_{C_{221013}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{11}lambda12Inv2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}g_1^2\overline{y_1y_2}}{8MF}$	$\frac{3\sqrt{11}lambda12Inv2\overline{y_1y_2}}{22MF}$

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1}y_2}{MF}$
T2	$\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y^e}}{MF}$
T3	$\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{11\sqrt{3}g_2^2\overline{y_1}y_2}{8MF}$	$-\frac{\sqrt{3}ye\overline{y_1}y_2\overline{y^e}}{12MF}$	$\frac{\sqrt{3}ig_2^2\overline{y_1}y_2}{6MF}$

$$16\pi^2\mu\frac{dC_{221014}}{d\mu}=\beta_{C_{221014}}^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{MF}$
T2	$\frac{\sqrt{3}i\lambda da12Inv1\overline{y_1y_2}}{MF}$
T3	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}g_2^2\overline{y_1y_2}}{6MF}$	$-\frac{\sqrt{3}i\lambda da12Inv1\overline{y_1y_2}}{6MF}$	$\frac{\sqrt{3}ig_1^2\overline{y_1y_2}}{8MF}$

Quantity: 2, 2, 10, 14

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{11} i \lambda d a 12 I n v 2 y_1 y_2}{M F}$
T2	$\frac{\sqrt{3} i g_2^2 y_1 y_2}{M F}$
T3	$\frac{\sqrt{3} i y e \overline{y_1 y_2 y e}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{3 \sqrt{11} i \lambda d a 12 I n v 2 y_1 y_2}{22 M F}$	$\frac{11 \sqrt{3} i g_2^2 y_1 y_2}{8 M F}$	$- \frac{\sqrt{3} i y e \overline{y_1 y_2 y e}}{12 M F}$